

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$
 $\frac{\langle a, b \mid ab = ba \rangle}{\mathbb{Z} \times \mathbb{Z}}$
 $S^1 \vee S^1$
 $\frac{S^1 \vee S^1}{\mathbb{Z} * \mathbb{Z}}$
 $T = \mathbb{Z} \times \mathbb{Z}$
 $T \hookrightarrow \mathbb{Z} * \mathbb{Z}$
 $f: \mathbb{Z} * \mathbb{Z} \rightarrow \mathbb{Z} * \mathbb{Z}$

[illegible][illegible]

Handwritten notes on graph paper:

- Left side: $\frac{1}{2} \frac{1}{2} \frac{1}{2}$
- Middle: $\frac{1}{2}$
- Right side: $\frac{1}{2}$